

Singularity Formation in Burgers' Equation

Lloyd Sangwoo Ko

Spring 2025 Applied Mathematics Directed Reading Program

Abstract

We study Burgers' equation with smooth initial data, focusing on the construction of smooth solutions via the method of characteristics. In particular, we analyze the development of potential singularities and the blow-up behavior arising from specific initial conditions. We also explore the existence of self-similar blow-up solutions and use them to describe the local structure of a gradient catastrophe.

1. Introduction

The inviscid Burgers' equation

$$u_t + u u_x = 0, \quad u(0, x) = u_0(x),$$

is a fundamental first-order quasilinear partial differential equation that models wave steepening and nonlinear advection. Despite the simplicity of the equation, smooth initial data need not produce smooth solutions for all time. Instead, characteristics may collide, causing the spatial derivative of the solution to become singular in finite time even while the solution itself remains bounded.

The purpose of this report is to study the structure and breakdown of classical solutions to the inviscid Burgers' equation. We focus on two related questions. First, how can classical solutions be constructed using the method of characteristics, and under what conditions does this construction fail? Second, what mechanism leads to finite-time blow-up, and how can the local behavior near the singularity be described through a self-similar structure?

We begin by deriving the characteristic flow associated with Burgers' equation and identifying the condition under which that flow ceases to be invertible. This provides a direct criterion for shock formation. We then introduce a self-similar ansatz near a singular time T and derive an ordinary differential equation for the corresponding blow-up profile. The resulting family of profiles illustrates how a bounded solution can nevertheless develop an unbounded spatial gradient.

2. Burgers' Equation and the Method of Characteristics

Consider the initial value problem

$$u_t + u u_x = 0, \quad u(0, x) = u_0(x).$$

Let $\phi_t(x)$ denote the characteristic curve emanating from x at time $t = 0$, defined by

$$\begin{cases} \dot{\phi}_t(x) = u(t, \phi_t(x)), \\ \phi_0(x) = x. \end{cases}$$

Along such a trajectory, the chain rule gives

$$\frac{d}{dt}u(t, \phi_t(x)) = u_t(t, \phi_t(x)) + u_x(t, \phi_t(x))\dot{\phi}_t(x) = u_t + u u_x = 0.$$

Thus

$$u(t, \phi_t(x)) = u_0(x).$$

Substituting this identity back into the characteristic equation gives

$$\dot{\phi}_t(x) = u_0(x),$$

and integration in time yields

$$\phi_t(x) = x + t u_0(x).$$

This formula gives a geometric interpretation of the solution. Each initial point x moves with constant velocity $u_0(x)$. As long as ϕ_t remains invertible, the classical solution is implicitly given by

$$u(t, x) = u_0(\phi_t^{-1}(x)).$$

3. Formation of Shocks

The characteristic representation remains valid only while the flow map ϕ_t is invertible. Differentiating with respect to the initial position gives

$$\phi'_t(x) = 1 + t u'_0(x).$$

By the Inverse Function Theorem, ϕ_t is locally invertible near x whenever

$$\phi'_t(x) \neq 0.$$

Thus the characteristic construction begins to break down when

$$1 + t u'_0(x) = 0.$$

Geometrically, this condition corresponds to the collision of characteristics. Two distinct trajectories can reach the same spatial point, after which the classical characteristic description no longer defines a single-valued smooth solution. This is the onset of a gradient catastrophe, or shock formation.

If u'_0 attains a negative minimum, the first singular time is

$$T = -\frac{1}{\min_x u'_0(x)}.$$

Therefore, if $u'_0(x) < 0$ somewhere, a shock forms in finite positive time.

The mechanism can also be seen directly in the derivative. Along the characteristic flow,

$$u_x(t, \phi_t(x)) = \frac{u'_0(x)}{1 + t u'_0(x)}.$$

When $u'_0(x) < 0$, the denominator approaches zero at the corresponding breakdown time, and the spatial derivative becomes singular while the transported value of u remains bounded.

4. Self-Similar Blow-Up

To investigate the local structure of the singularity near a blow-up time T , we introduce the self-similar ansatz

$$x = \lambda(t)y, \quad u(t, x) = \nu(t)Q\left(\frac{x}{\lambda(t)}\right),$$

where

$$\lambda(t) = (T - t)^b$$

for some exponent $b > 0$. Choosing

$$\nu(t) = \frac{\lambda(t)}{T - t} = (T - t)^{b-1},$$

and substituting the ansatz into

$$u_t + u u_x = 0$$

reduces the PDE to the ordinary differential equation

$$byQ' - (b - 1)Q + QQ' = 0.$$

Thus the study of self-similar singularities is reduced to understanding solutions of this profile equation.

5. Self-Similar Profiles

The reduced equation admits a family of smooth, odd profiles $Q_{k,a}$, parameterized by $k \in \mathbb{N}_0$ and $a > 0$, satisfying the implicit relation

$$y = -Q_{k,a} - aQ_{k,a}^{2k+1},$$

with

$$b = \frac{2k+1}{2k}.$$

These profiles are smooth and odd and satisfy

$$Q'_{k,a}(y) \geq -1.$$

Near $y = 0$, the poster gives the expansion

$$Q_{k,a}(y) = -y - ay^{2k+1} + O(|y|^{2k+2}).$$

In particular,

$$Q'_{k,a}(0) = -1.$$

As $y \rightarrow \pm\infty$, the profiles have the asymptotic behavior

$$Q_{k,a}(y) \sim \mp \left(\frac{|y|}{a} \right)^{1/(2k+1)}.$$

Returning to the original variables gives the blow-up profile

$$u(t, x) = (T-t)^{b-1} Q_{k,a} \left(\frac{x}{(T-t)^b} \right).$$

This representation separates the shrinking spatial scale $(T-t)^b$ from the amplitude scale $(T-t)^{b-1}$ and describes the solution close to the developing singularity.

6. Gradient Blow-Up

Differentiating the self-similar form with respect to x gives

$$u_x(t, x) = (T-t)^{-1} Q'_{k,a} \left(\frac{x}{(T-t)^b} \right).$$

At the origin,

$$u_x(t, 0) = (T-t)^{-1} Q'_{k,a}(0).$$

Since $Q'_{k,a}(0) = -1$,

$$u_x(t, 0) = -\frac{1}{T-t} \longrightarrow -\infty \quad \text{as } t \rightarrow T^-.$$

Thus the singularity is a gradient catastrophe: the solution itself remains bounded while its spatial derivative diverges. This agrees with the characteristic description. In the characteristic picture, trajectories become compressed until the flow map loses invertibility. In the self-similar picture, the same phenomenon appears as a $(T-t)^{-1}$ divergence of the spatial gradient.

The profiles $Q_{k,a}$ provide additional information about the local structure near the singularity. Rather than merely identifying the time at which the classical solution fails, the self-similar formulation describes how the solution approaches that failure under a rescaling of space and amplitude.

7. Conclusion

The inviscid Burgers' equation provides a simple setting in which smooth solutions can develop finite-time singularities. The method of characteristics gives the explicit flow map

$$\phi_t(x) = x + t u_0(x),$$

and loss of invertibility of this map determines the onset of shock formation. In particular, negative values of u'_0 cause characteristics to collide, with the first singular time determined by the most negative initial slope.

The self-similar formulation complements this characteristic picture by describing the local structure near the singularity. After introducing an appropriate shrinking spatial scale, Burgers' equation reduces to an ODE for a family of profiles $Q_{k,a}$. These profiles show explicitly how the solution can remain bounded while its spatial derivative diverges like $(T - t)^{-1}$.

Together, the characteristic and self-similar approaches provide two perspectives on the same gradient catastrophe: one describes the geometric mechanism leading to breakdown, while the other describes the local structure of the solution as the singular time is approached.